

Report 3: Full panel gallery, quantization panels, paper-curve digitization

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Enhancing Adaptive Deep Brain Stimulation via Efficient Reinforcement Learning (Mehregan et al.)
Closed-Loop Neuromodulator Deep Brain Stimulation (Nguyen et al.)
Sample-Efficient Reinforcement Learning Controller for Deep Brain Stimulation in Parkinson's Disease (Ravivarapu et al.)

Overview (August 10, 2026): Mehregan: Fig. 1b–6b **Pass** on current automated gates; Fig. 4a–4b flagged for revisit since the digitizations. Nguyen: Fig. 3 **Pass**; Fig. 4 **in progress** (reward-plateau shape gates). Ravivarapu: Fig. 4a **in progress** (shape gates still fail). Process upgrade since Report 2: **paper-curve digitization**: quantify against refined digitized curves instead of eyeballing PNGs.

Same caveats as before: replications from paper text and figures (no released training / environment source), shared plant = Python reconstruction of **Kumaravelu et al. (2016)** cortex–basal ganglia–thalamus biophysical model (6-OHDA lesioned rat, *J. Comput. Neurosci.*). Full gates and run commands: [figures/mehregan/replications.md](#), [figures/nguyen/replications.md](#), [figures/ravivarapu/replications.md](#).

Paper-curve digitization (new since Report 2)

Why

Earlier reports judged “close enough” mostly by eye: ordering, onset timing, training shape. That works for coarse claims, but it fails when the question is *how suppressed, how gradual, or how high* a curve sits relative to the paper.

I now **digitize the paper curves into numeric series** and use those as the measurement ruler for gates and iteration.

Workflow (short)

- PIL color-mask hypothesis**: fast automated trace from the paper crop (*scripts/digitization/extract_pil.py* + a per-panel axis/color config).
- Seed WebPlotDigitizer**: turn the PIL points into a one-click .tar project (*pil_to_wpd.py*) so a human opens WPD with image + points already loaded.
- Human cleanup in WPD**: fix missed peaks, legend pollution, axis calibration; save a **refined project**.
- Normalize**: *normalize_wpd.py* reads the refined WPD **project JSON** (or a WPD CSV export), pulls each series **already-calibrated** (x, y) pairs from WPD’s stored value fields (it does **not** remap pixels through axes again), optionally renames legend labels to canonical keys (e.g. *pd*, *pd_130hz*), and writes *curves_wpd_refined*.json*: per series, n , full xy in paper data units, and *color_rgba* when WPD has it. No baked-in early/late means; gates slice xy by real x -windows later.
- Gate & overlay**: *panel_plot.py* loads the refined JSON, computes **x -window** means / ratios / drops, and draws a dashed **Paper (digitized)** overlay next to my solid replication traces.

How it improves iteration

Before	After
"Looks a bit low at the end"	Late-window mean ratio vs digitized paper (pass/fail with a number)
Tune until the PNG <i>feels</i> right	Abort early when a digitized ratio is already impossible
Vague pass/fail from eyeballing	Gates use the same numeric series as overlays on replication plots
Vague feedback across seeds	Same x -windows and series names for every replot

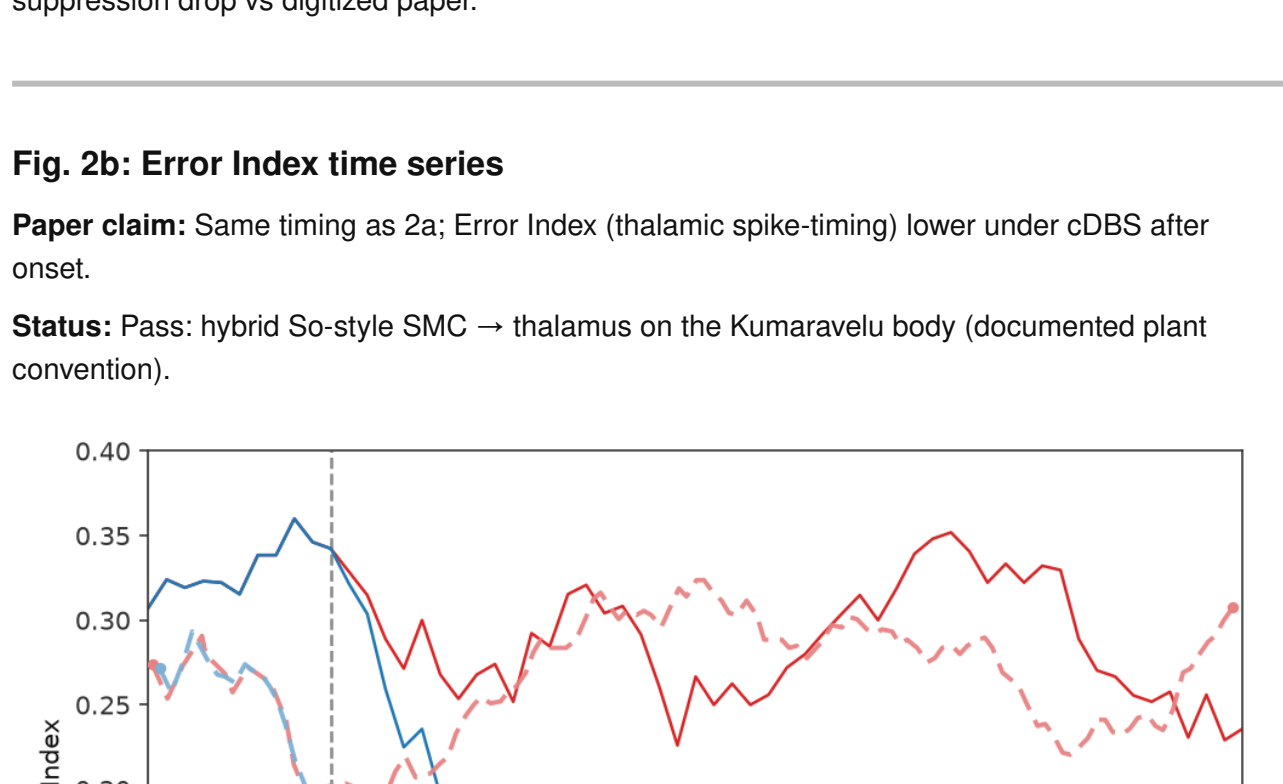
All gallery panels below have refined WPD anchors in each panel’s *paper_digitization/* folder (Mehregan: *curves_wpd_refined*.json*; Nguyen and Ravivarapu: *normalized_curves_fig*.json* from refined WPD exports). Replication plots draw dashed **Paper (digitized)** overlays on the same axes during tuning.

Gallery: Mehregan et al.

Fig. 1b: GPI power spectral density

Paper claim: PD elevates GPI beta vs healthy; 130 Hz cDBS suppresses that elevation.

Status: Pass: ordering and beta-peak shape match (seeds 0–9 mean).



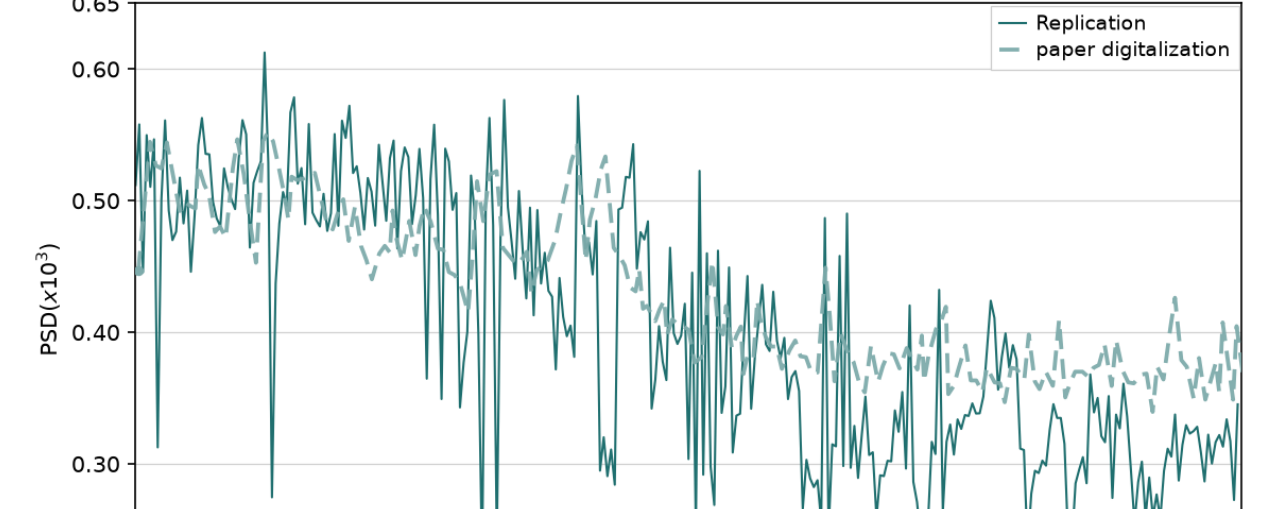
"Oscillatory activity of model neurons in the GPI under the healthy state, PD state, and PD state with 130Hz conventional stimulation." *Mehregan et al.*

Gates example: PD > healthy on beta; 130 Hz cDBS < untreated PD; suppression ratio and beta levels within band of digitized paper.

Fig. 2a: GPI beta power time series

Paper claim: After cDBS onset at 2 s, treated (blue) falls and stays below untreated PD (red); shared pre-stim baseline.

Status: Pass: blue-below-red after $t=2$, shared 0–2 s baseline, dense trailing protocol.



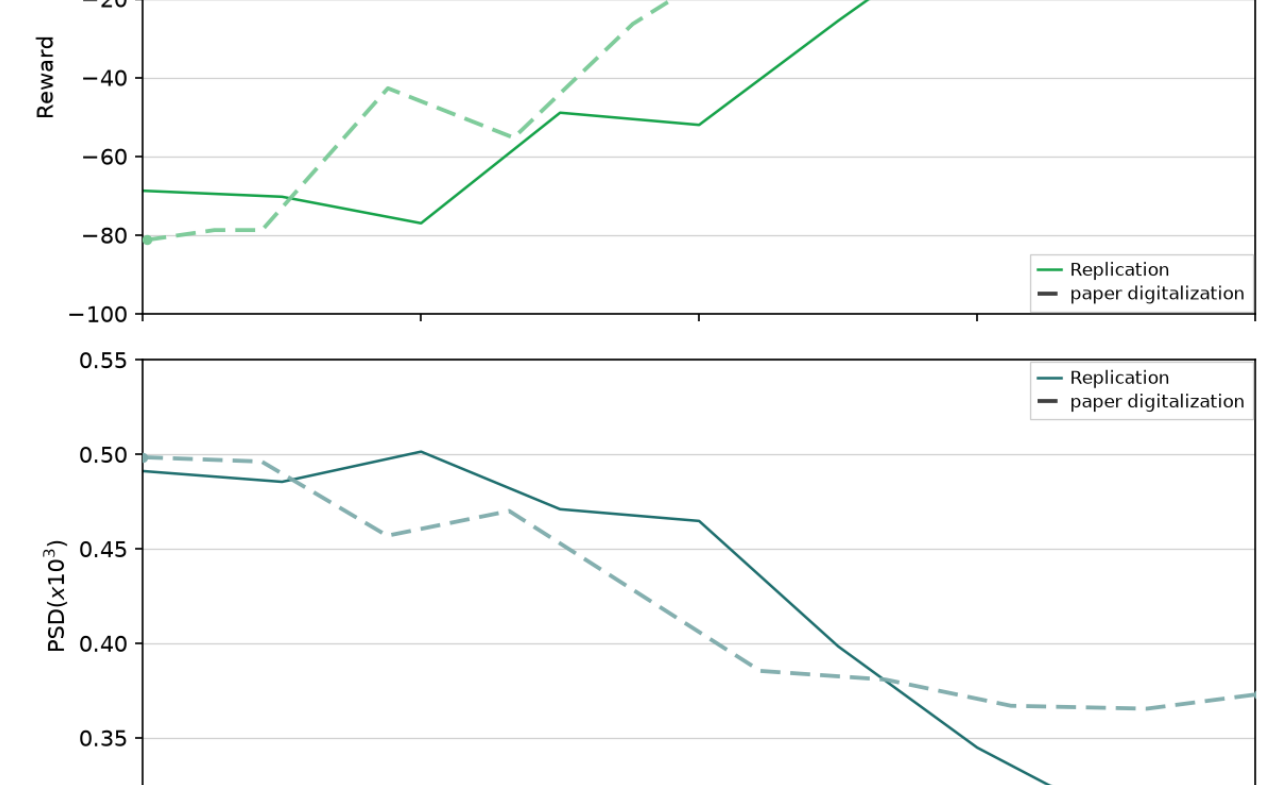
"cDBS effects on beta power of GPI neurons" *Mehregan et al.*

Gates example: Shared pre-onset baseline; treated < untreated after onset; late ratio and suppression drop vs digitized paper.

Fig. 2b: Error Index time series

Paper claim: Same timing as 2a; Error Index (thalamic spike-timing) lower under cDBS after onset.

Status: Pass: hybrid So-style SMC → thalamus on the Kumaravelu body (documented plant convention).



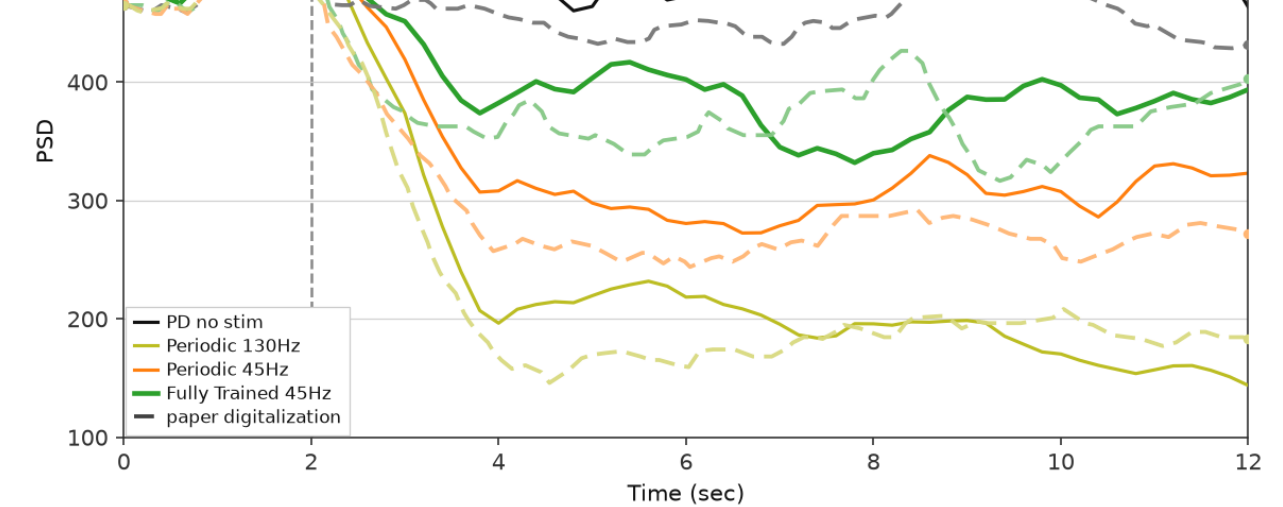
"cDBS effects on error." *Mehregan et al.*

Gates example: Same timing gates as Fig. 2a on the Error Index biomarker (pre-onset, late ordering, digitization ratios).

Fig. 4a: Training beta vs step

Paper claim: Noisy early β , sharp mid-run drop, lower late plateau (45 Hz DDPG training).

Status: Revisit in progress: pair with 4b; softmax τ 3→1.4 to avoid pattern-0 collapse (locked v18 was $\tau=1.0$).



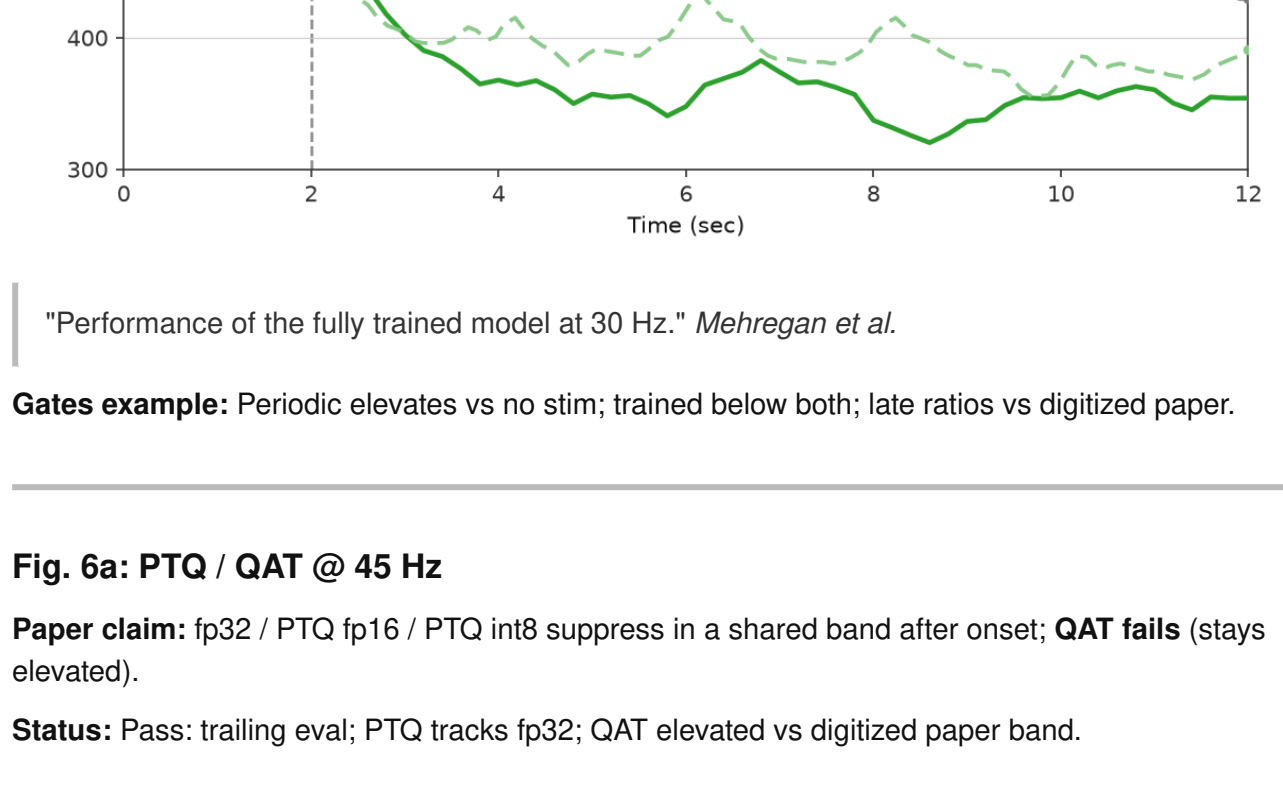
"The average power of the beta frequency band for each step during training of the 45Hz average frequency model." *Mehregan et al.*

Gates example: Training trend down; mid-run fade and late/early ratio vs digitized paper; drop magnitude $\geq 70\%$ of paper drop.

Fig. 4b: Training reward and episode-mean beta

Paper claim: Over episodes 0–8, reward rises toward zero while episode-mean β falls.

Status: Revisit in progress: match digitized late PSD floor (~ 0.37 , above β_1) so reward approaches 0 from below; paired with latest Fig. 4a series.



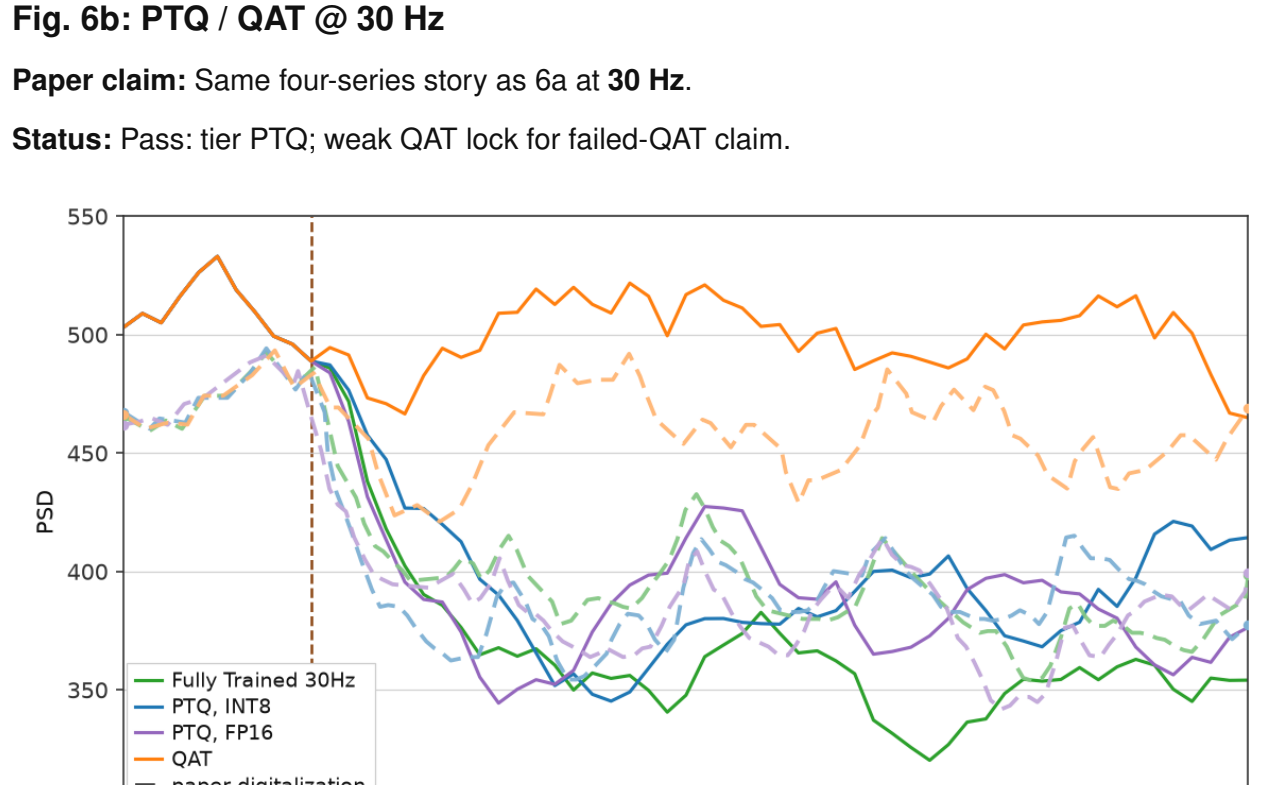
"Beta power in GPI during each episode of training of the 45Hz (top) and accumulated reward of each training episode for the 45Hz (bottom)." *Mehregan et al.*

Gates example: Reward rises from negative early values; episode-mean PSD falls; rise timing and PSD drop ratio vs digitized paper; late PSD ≥ 0.35 and within 15% of paper; late reward in $(-10, 2]$.

Fig. 5a: Post-train efficacy @ 45 Hz

Paper claim: Trained irregular pattern suppresses β vs no stim, but stays above periodic 45 Hz; 130 Hz lowest.

Status: Pass: skip_regular alphabet; four-series trailing eval.



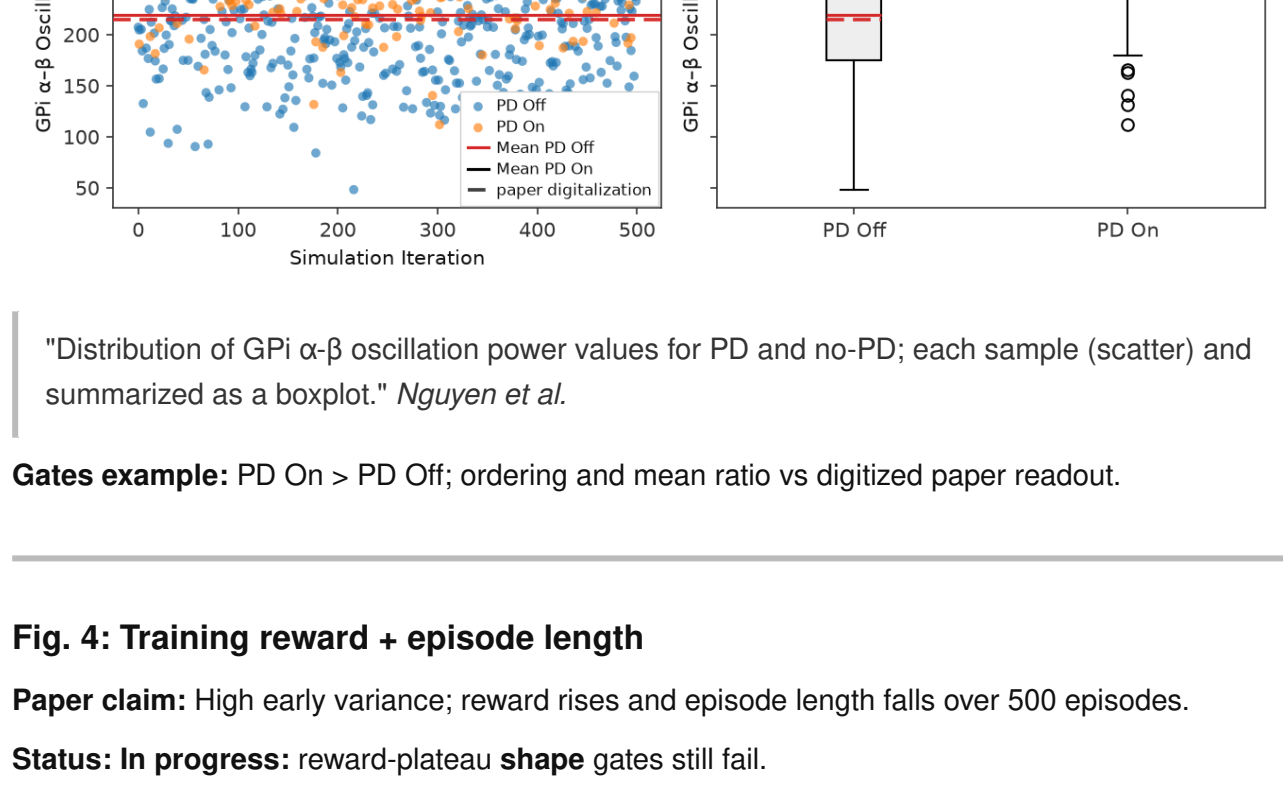
"Performance of the fully trained model at 45 Hz." *Mehregan et al.*

Gates example: Shared baseline; ordering 130 Hz < trained < no stim < periodic 45 Hz; late trained/no-stim and periodic/no-stim ratios vs digitized paper.

Fig. 5b: Post-train efficacy @ 30 Hz

Paper claim: Periodic 30 Hz *elevates* β ; trained irregular pattern goes *below* both no stim and periodic.

Status: Pass: burst-pattern alphabet (mean rate still 30 Hz).



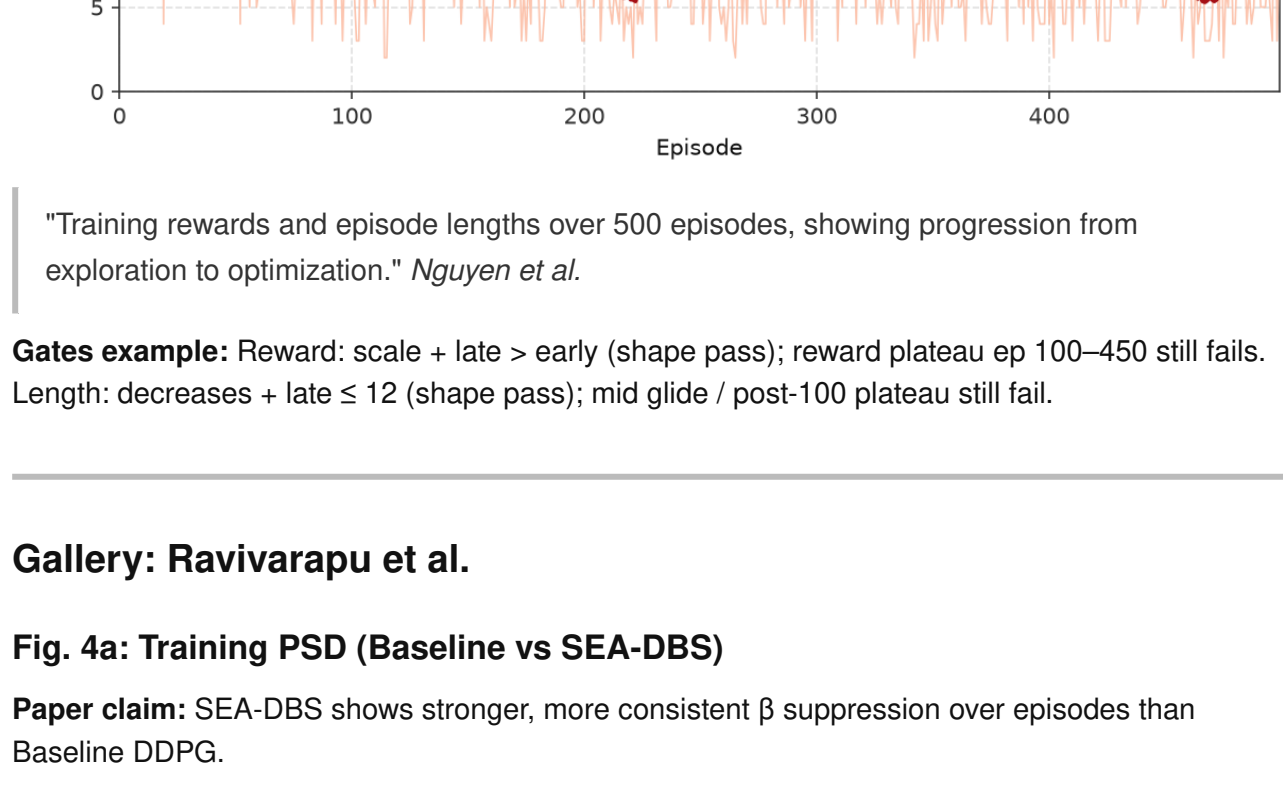
"Performance of the fully trained model at 30 Hz." *Mehregan et al.*

Gates example: Periodic elevates vs no stim; trained below both; late ratios vs digitized paper.

Fig. 6a: PTQ / QAT @ 45 Hz

Paper claim: fp32 / PTQ fp16 / PTQ int8 *suppress* in a shared band after onset; **QAT fails** (stays elevated).

Status: Pass: trailing eval; PTQ tracks fp32; QAT elevated vs digitized paper band.



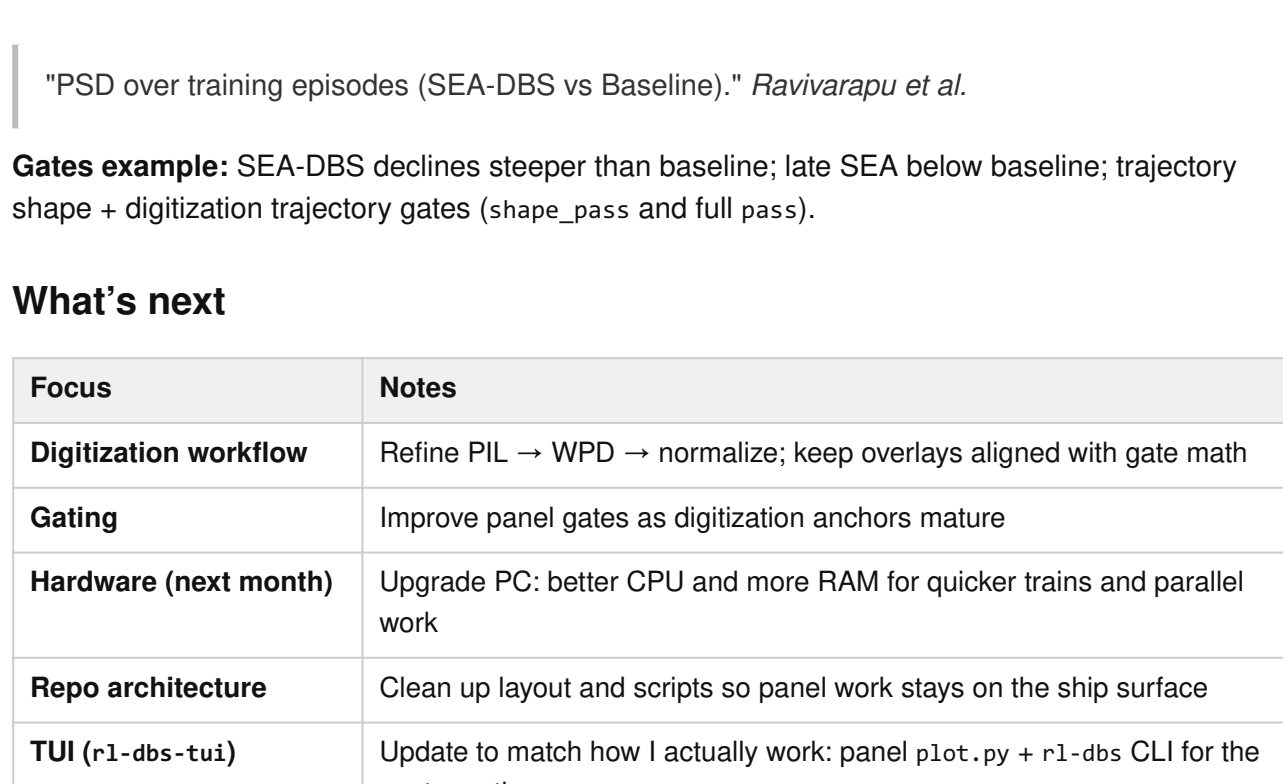
"Performance PTQ and QAT at 45 Hz." *Mehregan et al.*

Gates example: fp32 suppresses; PTQ fp16/int8 track fp32 and stay distinct; QAT elevated vs fp32 and near baseline; digitization-banded level gates for all four series.

Fig. 6b: PTQ / QAT @ 30 Hz

Paper claim: Same four-series story as 6a at 30 Hz.

Status: Pass: tier PTQ; weak QAT lock for failed-QAT claim.



"Performance PTQ and QAT at 30 Hz." *Mehregan et al.*

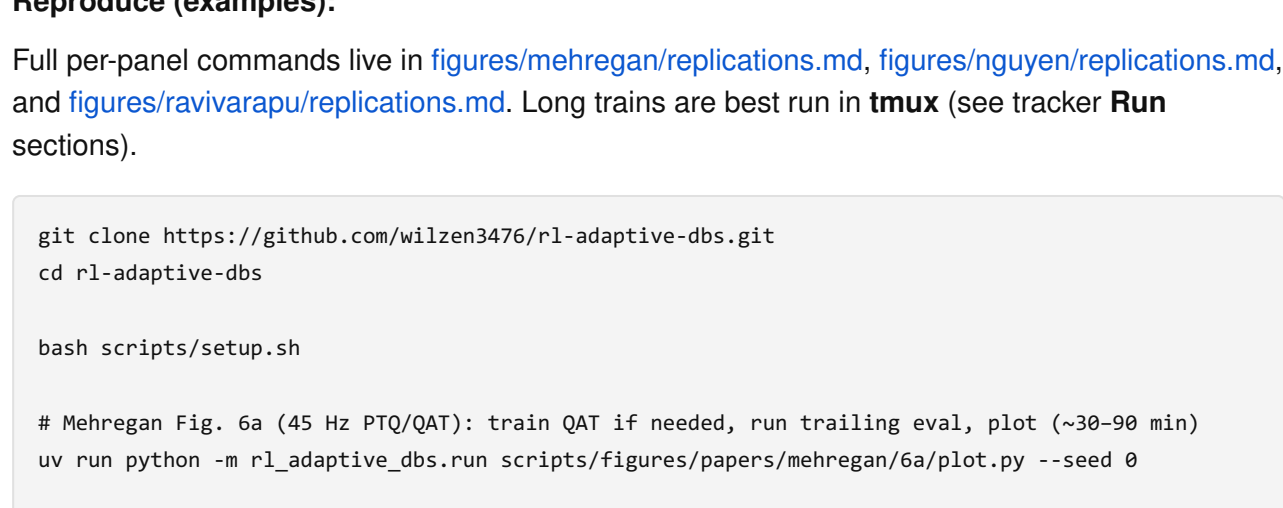
Gates example: Same qualitative split as 6a at 30 Hz: PTQ tracks fp32; QAT elevated; digitization level gates.

Gallery: Nguyen et al.

Fig. 3: GPI α - β distribution

Paper claim: PD On α - β power higher than PD Off.

Status: Pass: 500 $\times$ 100 ms samples; 7–35 Hz α - β index (not Mehregan’s 13–35 Hz P β).



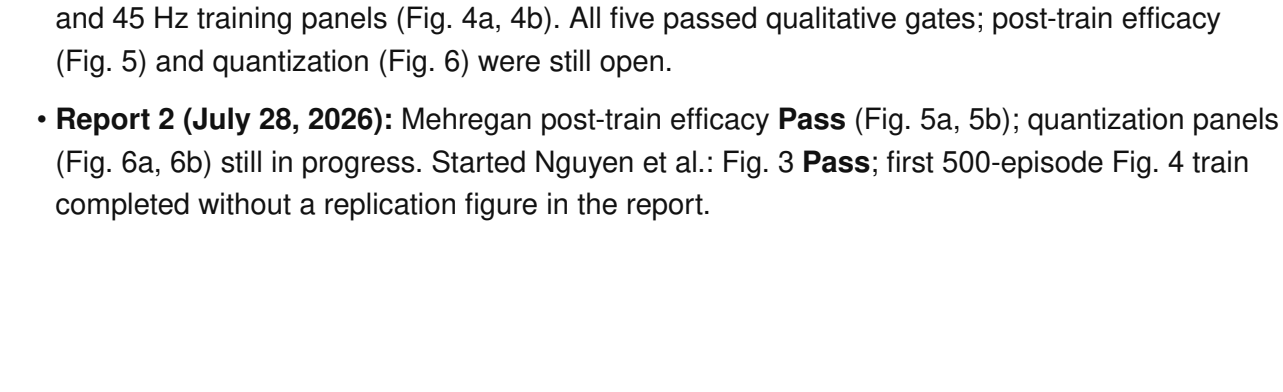
"Distribution of GPI α - β oscillation power values for PD and no-PD; each sample (scatter) and summarized as a box plot." *Nguyen et al.*

Gates example: PD On > PD Off; ordering and mean ratio vs digitized paper readout.

Fig. 4: Training reward + episode length

Paper claim: High early variance; reward rises and episode length falls over 500 episodes.

Status: In progress: reward-plateau *shape* gates still fail.



"Training rewards and episode lengths over 500 episodes, showing progression from exploration to optimization." *Nguyen et al.*

Gates example: Reward: scale + late > early (shape pass); reward plateau ep 100–450 still fails. Length: decreases + late ≤ 12 (shape pass); mid glide / post-100 plateau still fail.

Gallery: Ravivarapu et al.

Fig. 4a: Training PSD (Baseline vs SEA-DBS)

Paper claim: SEA-DBS shows stronger, more consistent β suppression over episodes than Baseline DDPG.

Status: In progress: *shape* gates still fail; digitized paper overlay on the same axes.

"PSD over training episodes (SEA-DBS vs Baseline)." *Ravivarapu et al.*

Gates example: SEA-DBS declines steeper than baseline; late SEA below baseline; trajectory shape + digitization trajectory gates (*shape_pass* and full pass).

What’s next

Focus	Notes
Digitization workflow	Refine PIL → WPD → normalize; keep overlays aligned with gate math
Gating	Improve panel gates as digitization anchors mature
Hardware (next month)	Upgrade PC: better CPU and more RAM for quicker trains and parallel work
Repo architecture	Clean up layout and scripts so panel work stays on the ship surface
TUI (r1-dbs-tui)	Update to match how I actually work: <i>panel_plot.py</i> + <i>r1-dbs</i> CLI for the past month
Docs	Keep docs/ and figure trackers aligned with current behavior and gates
Setup verification	Re-run scripts/setup.sh on a clean machine and fix anything that breaks
Nguyen Fig. 4 (p2f4)	Close reward/length <i>shape</i> gates (in progress)
Ravivarapu Fig. 4a (p3f4a)	Finish SEA-DBS vs Baseline training panel (in progress)
Nguyen Fig. 5–7	Remaining Nguyen panels after Fig. 4
Ravivarapu Fig. 4b–7	Remaining Ravivarapu panels after Fig. 4a
Mehregan Fig. 4a–4b	Revisit since the digitizations (Pass on gates today)

How this was produced (delta since Report 2)

- Digitization stack**: PIL → WPD seed → refined normalize → *paper_gates* / panel overlays across all three papers.
- Mehregan 6a/6b**: Pass on digitization-backed quantization gates.
- Nguyen**: Fig. 3 **Pass**; Fig. 4 **in progress** (reward/length shape gates).
- Ravivarapu**: Fig. 4a **in progress** (shape gates still fail).

Reproduce (examples):

Full per-panel commands live in [figures/mehregan/replications.md](#), [figures/nguyen/replications.md](#), and [figures/ravivarapu/replications.md](#). Long trains are best run in **tmux** (see tracker **Run** sections).

```
git clone https://github.com/wilzen3476/r1-adaptive-dbs.git
cd r1-adaptive-dbs

bash scripts/setup.sh

# Mehregan Fig. 6a (45 Hz PTQ/QAT): train QAT if needed, run trailing eval, plot (~30–90 min)
uv run python -m r1_adaptive_dbs.run scripts/figures/papers/mehregan/6a/plot.py --seed 0

# Replot from cached eval.json (no train/eval)
uv run python -m r1_adaptive_dbs.run scripts/figures/papers/mehregan/6a/plot.py --plot-only

# Nguyen Fig. 4: 500-episode DSON train + plot (long run)
uv run python -m r1_adaptive_dbs.run scripts/figures/papers/nguyen/4/plot.py

# Replot from cached training series (no train)
uv run python -m r1_adaptive_dbs.run scripts/figures/papers/nguyen/4/plot.py --plot-only

# Ravivarapu Fig. 4a: 150-episode Baseline vs SEA-DBS train + plot
uv run python -m r1_adaptive_dbs.run scripts/figures/papers/ravivarapu/4a/plot.py

# Replot from cached series / checkpoints (no train)
uv run python -m r1_adaptive_dbs.run scripts/figures/papers/ravivarapu/4a/plot.py --plot-only
```

Environment: All runs were executed and verified locally using *uv*. Setup instructions and dependencies are documented in *scripts/setup.sh* and the repository *README*.

Prior reports

• **Report 1 (July 14, 2026)**: First showcase for Mehregan et al.: plant panels (Fig. 1b, 2a, 2b) and 45 Hz training panels (Fig. 4a, 4b). All five passed qualitative gates; post-train efficacy (Fig. 5) and quantization (Fig. 6) were still open.

• **Report 2 (July 28, 2026)**: Mehregan post-train efficacy **Pass** (Fig. 5a, 5b); quantization panels (Fig. 6a, 6b) still in progress. Started Nguyen et al.: Fig. 3 **Pass**; first 500-episode Fig. 4 train completed without a replication figure in the report.